SMART INDIA HACKATHON 2026

- **Problem Statement ID – 26176**
- **Problem Statement Title - ORCA Marine EcoSystem Reasoning with Collaborative Agents**
- **Theme - Space Technology**
- **PS Category - Software**
- **Team ID - 175766**
- **Team Name (Registered on portal) : MARINES**

MARVO

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Challenges and Problems

- **Fragmented Marine Intelligence -** 535 fishermen arrested in 2024 for inadvertently crossing the IMBL — ISRO, IMD, and INCOIS data existed but no system synthesized it into one decision.
- **Last-Mile Advisory Gap -** INCOIS advisories reach 52% of fishermen, but only 35% use them — because chlorophyll maps are hard to read on small screens at 4 AM.
- **No Unified Decision Layer -** 400 fishermen die annually on Indian coasts — yet no single system combines risk, fishing zones, routes, and travel time into one clear answer.

Innovation & Uniqueness

- **Proactive, Not Reactive -** Pre-departure Drift/SAR Envelope + IMBL Current Guard identify potential drift and boundary-crossing risks before an emergency occurs.
- **Peer-Reviewed Fishing Intelligence -** Fishing zones computed using the Habitat Suitability Index methodology (Sarangi et al., CMFRI 2024) - the same method INCOIS uses, now voice-delivered.
- **AI Never Controls Safety Math -** No model output, user input, or override can bypass certified safety floors. When IMD says "do not venture," ORCA locks at UNSAFE. Always.

Proposed Solution

MARVO - Agentic Marine Intelligence Platform

ORCA synthesizes fragmented marine data through 10 collaborative AI agents to generate safe , explainable and location-specific decisions - with evidence, safety overrides and multilingual delivery.

For Fishermen

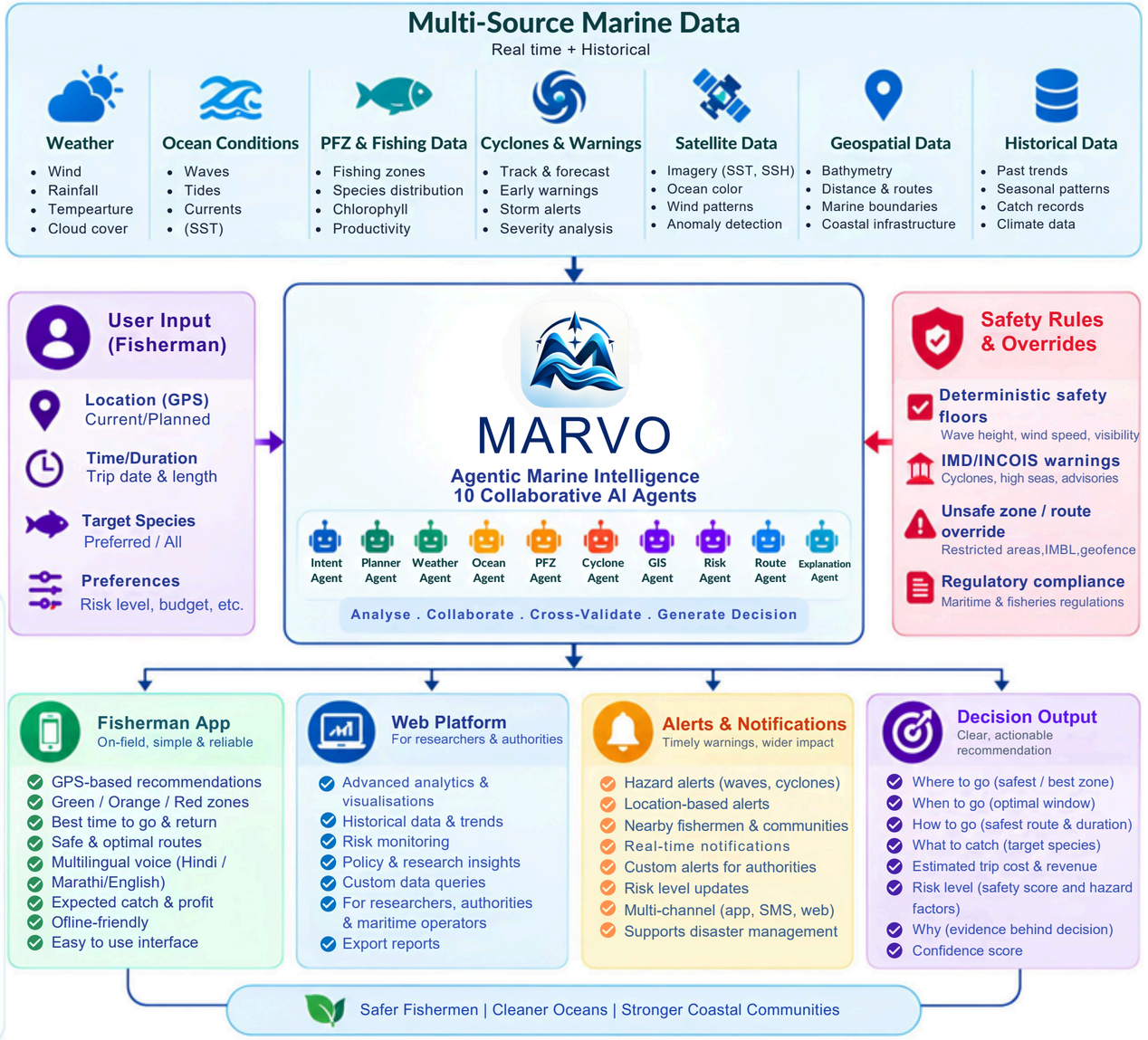
- GPS-based fishing zone recommendations
- Risk score + GO/CAUTION/NO-GO
- Best time and route to fish
- Species-specific fishing probability
- Return-before risk guidance
- IMBL drift alerts
- Offline first access and multilingual service

For Safety & Disaster Management

- Real time hazard and cyclone alerts
- Vessels-at-risk during projected paths
- Pre-computed SAR drift envelopes
- Location-based mass alerts
- Restricted-zone and IMBL monitoring
- Support for coastal authorities
- Unified marine risk view

For Researchers & Authorities

- Evidence-backed marine intelligence
- Historical trends & anamoly analysis
- PFZ validation through catch reports
- Policy and research insights
- Vessel registory & geofence violation
- Custom data queries & visualisations
- Future expansion to blue economy use cases



TECHNICAL APPROACH

From Data to Decisions | Multi - Agent Intelligence | Real Impact

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1. TECHNOLOGIES TO BE USED

Frontend (User Interface)

React 18 | TypeScript | Tailwind CSS | Mapbox | Vite

Backend (Core Engine)

Python 3.10 | FastAPI | Pydantic v2 | Lang Graph | HTTPX

Key Libraries & Tools

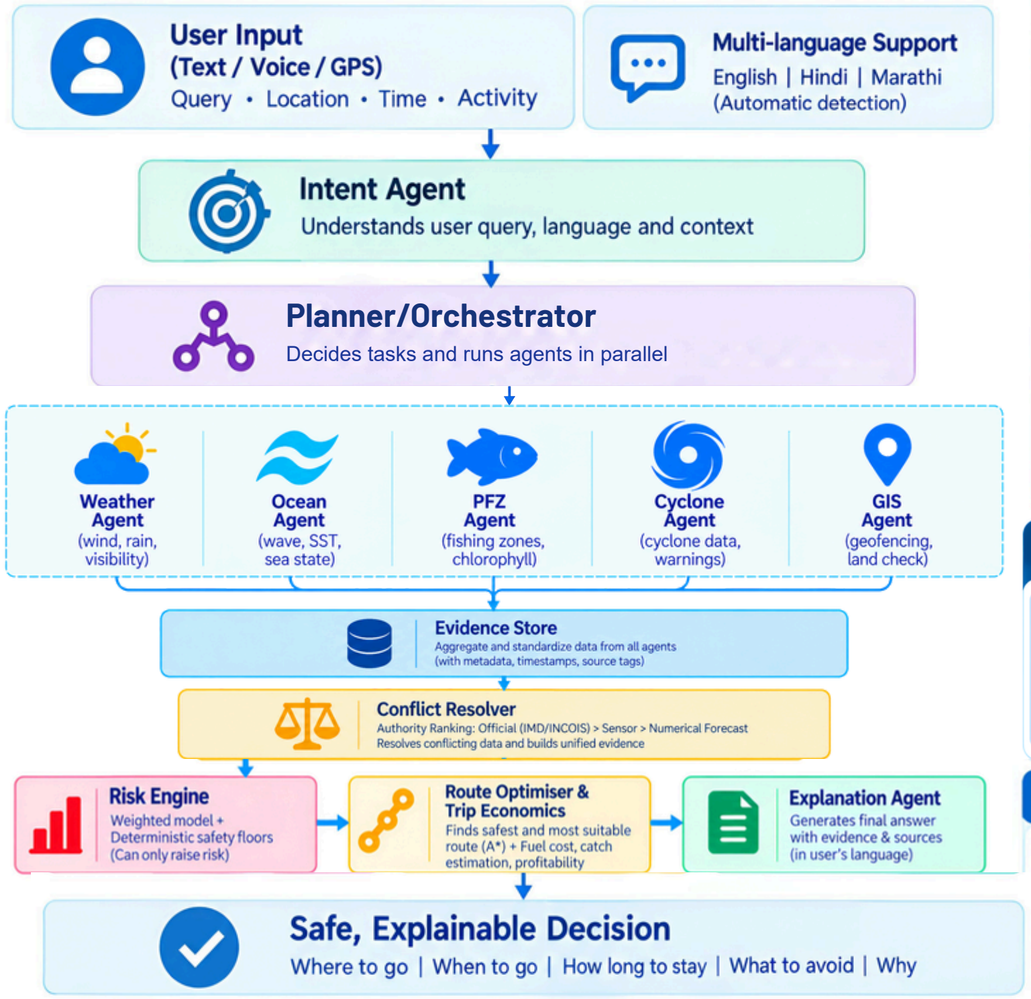
- 10-Agent Architecture
- GIS & Geofencing
- Deterministic Risk Engine
- Route Optimisation (A*)
- SARVAM AI (STT + TTS)
- In Memory Cache (72+ hr Series)
- Geospatial Analysis
- Offline Mode (PWA)

Hardware (for Development & Deployment)

- Standard Laptop/PC (Windows/Linux/Mac)
- Cloud Server (Optional, for deployment)

2. METHODOLOGY AND PROCESS FOR IMPLEMENTATION

Flowchart of MARVO's working



3. DATA SOURCES

Open-Meteo Marine	Wave height, wave period, SST	Live (keyless)
Open-Meteo Forecast	Wind, rain, visibility, air temperature	Live (keyless)
INCOIS (Ocean Data)	PFZ advisories, ocean state	Live (via data-sharing)
IMD (Weather & Cyclone)	Marine warnings, cyclone bulletins	Live (via bulletins)
ISRO MODDAC (Satellite)	Satellite SST / ocean colour	Live (via data-sharing)
GDACS (Global Disaster Alert)	Disaster alerts, cyclone & hazard data	Live
OBIS / Map of Life	Species occurrence data	Offline snapshot (bundled)
OpenStreetMap	Basemap, coastline, geofencing	Keyless

4. DATA PROCESSING PIPELINE

Data Collection (Multi-source) → Data Normalisation & Caching (72-hr series, 10-min TTL) → Conflict Resolution (Authority Ranking) → Model & Reasoning (Fishing model + Risk scoring) → Safety Overrides (Deterministic floors) → Route Optimisation (A*) → Trip Economics (Cost/Revenue Analysis) → Explainable Output (with sources)

5. WORKING PROTOTYPE

- Web Application (Full UI)
- PWA (Mobile) (Add to Home Screen)
- Demo Mode (Works Offline)
- Live Mode (Open-Meteo)

FEASIBILITY AND VIABILITY

Practical | Scalable | Sustainable | Real-World Impact

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01 TECHNICAL FEASIBILITY

Can ORCA be built with existing technology?

- Built using **proven and widely available technologies** (Python, FastAPI, React, etc.)
- Uses **open and government data sources** (IMD, INCOIS, Open-Meteo, GDACS, MOSDAC)
- **10-agent hybrid architecture —** LLM for language, deterministic for safety
- Runs on **standard devices** (mobile PWA + web) with offline support
- **Modular and extensible design** for future expansion

02 DATA & OPERATIONAL FEASIBILITY

- **Uses real time & historical marine data** (Open-Meteo, INCOIS, IMD, GDACS, MOSDAC)
- **Integrates multiple datasets:** waves, SST, chlorophyll, currents, tides, alerts
- **Location based analysis** using GPS & coastal grids
- **10-day forecast** calendar with tiered background data refresh
- **Fallback mechanisms** for data unavailability
- **Multilingual voice interface** (Hindi, Marathi, English)

03 ECONOMIC & SOCIAL VIABILITY

- Helps fishermen make **safer** and **more informed decisions**, reducing loss of life
- 2.3× higher catch per trip via PFZ guidance, reducing fuel costs
- Supports **community safety** through early warnings and geofencing of hazardous/restricted areas
- Improves access to information in **local languages,** bridging digital divide
- Benefits **coastal economy** by improving catch efficiency & resource sustainability

04 POTENTIAL CHALLENGES AND RISKS

- **Limited availability** & inconsistency of some marine datasets
- **Connectivity issues** in remote coastal areas
- **Approvals and access** requirements for IMD/INCOIS data
- **Uncertainity in forecasts** & model predictions (extreme weather alerts)
- Need for **high-quality GIS data** for accurate boundaries (restricted zones & marine protected areas)
- **Ongoing maintenance,** API costs and model updates

STRATEGIES FOR OVERCOMING THESE CHALLENGES

How will we address them ?

01 Multi-Source Data Integration
- Combine IMD, INCOIS, Open-Meteo, GDACS etc.
- Use fallback and confidence scoring

02 Offline & Low-Connectivity Support
- PWA with catched data
- VCSS satellite transponder path for at-sea alerts

03 Partnerships & Approvals
- Collaborate with IMD, INCOIS and authorities
- Use publicly available datasets where possible

04 Robust & Explainable Models
- Hybrid approach: deterministic safety floors + LLM
- Show clear reasoning for every recommendation

05 User Training & Awareness
- Onboarding sessions with local communities
- Simple, multilingual and intuitive interface

06 Scalable Architecture
- Modular design for new data sources and regions
- Designed for long-term deployment and maintenance



IMPACT AND BENEFITS

Safer People | Stronger Communities | A Sustainable Ocean | A Brighter Tomorrow

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01 FISHERMEN

Safer, Evidence-Backed Trip Decisions

- Fishing probability, risk, route and travel time in **one decision**
- Real-time **hazard alerts** (high waves, cyclones, restricted zones)
- **Multilingual voice interface** (Hindi, English, Marathi)
- **Offline-first access —** full trip cached before departure
- **IMBL Current Guard** warns before boundary drift occurs

02 COASTAL COMMUNITIES & DISASTER MANAGEMENT

More Prepared. More Resilient.

- **Proactive cyclone, high-wave** & current-risk awareness
- **Vessels-at-Risk** counter inside cyclone projected path
- Pre-departure SAR **envelope** for every active voyage
- **Mass alert** broadcast to all vessels in a zone

03 AUTHORITIES & RESEARCHERS

Data-Driven Decision Making.

- Unified marine evidence layer for monitoring and analysis
- **Traceable source, timestamp** & confidence for every insight
- **Conflict resolution** across data sources
- **Automated 5nm violation log** for enforcement records
- **PFZ Validation Module —** catch reports vs ORCA predictions

04 BLUE ECONOMY & SCALABILITY

Sustainable Growth. Wider Impact.

- Improved **fishing efficiency** and better resource utilisation
- **Reduced operational costs** for fishermen
- Supports **sustainable use** of marine resources
- **Modular platform scalable** to other coastal regions
- **VCSS-ready** — integrates with 49,000+ ISRO transponders

DATA → Multi-source marine data (Real-time + Historical)

EVIDENCE → AI-powered analysis with source & confidence

DECISION → Actionable, explainable recommendations

SAFER OUTCOMES → People, communities and a sustainable ocean

RESEARCH AND REFERENCES

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Key Research Papers

Potential Fishing Zone Forecasting using Satellite Data

Study on PFZ estimation using SST, chlorophyll and ocean colour data.

View Paper

Machine Learning for Marine Weather Prediction

ML models for wave height, wind speed and cyclone prediction.

View Paper

Early Warning Systems For Fishermen Safety

Framework for coastal hazard alerts and risk communication.

View Paper

Ocean Data Integration for Decision Support

Multi-source data fusion for marine decision systems.

View Paper

Socio-Economic Impact of PFZ on Fishermen

Impact analysis of PFZ guidance on fishing income and safety.

View Paper

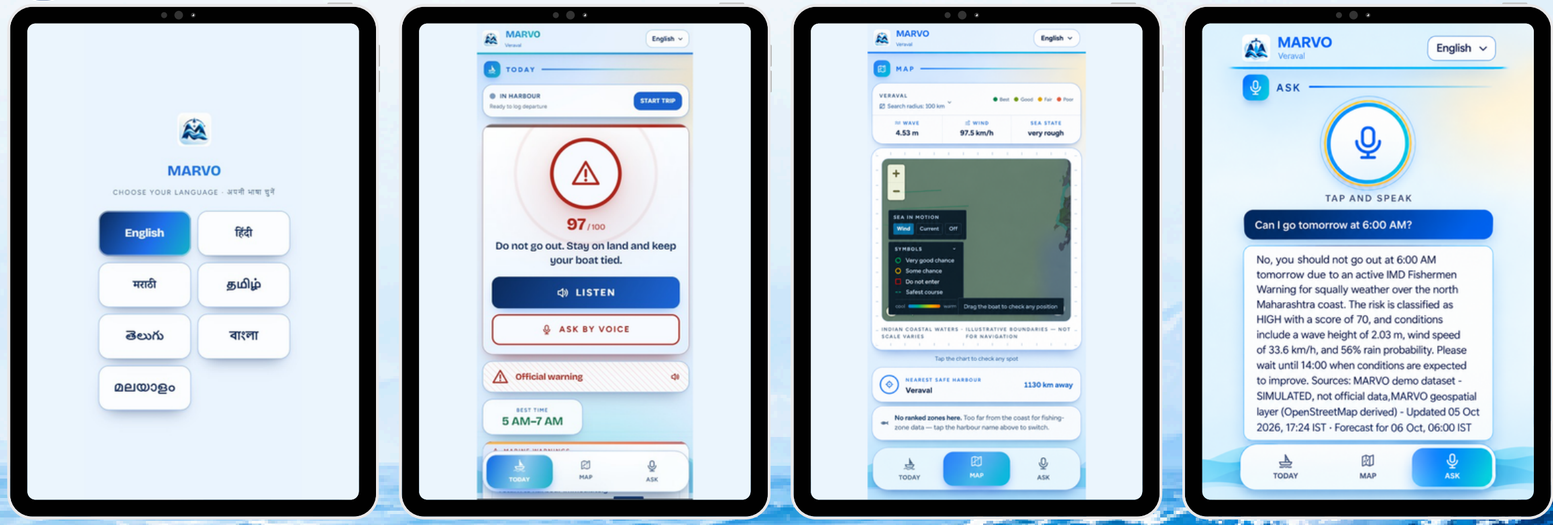
Authoritative Data Sources



Competitive Feature Matrix

Feature / Platform	SAMUDRA	Fisher Friend	NabhMitra	MARVO
PFZ / Fishing Zones	✓	✓	✓	✓
Weather & Ocean Forecasts	✓	✓	✓	✓
Cyclone / Hazard Alerts	✓	✓	✓	✓
GPS / Location Awareness	✓	✓	✓	✓
Navigation / Route Guidance	✓	✗	✓	✓
Offline / Low Connectivity	✗	✗	✓	✓
Geofencing / Restricted Zones	✗	✓	✓	✓
Expected Catch / Profit Info	✗	Limited (market info)	✗	✓
Species-Specific Probability	✗	✗	✓	✓
Risk + Fishing in One Decision	✗	✗	✓	✓
Multi-Source Data Fusion	✓	✗	✗	✓
Multi-Agent Reasoning	✗	✗	✗	✓
Evidence-Baked Explanation	✗	✗	✗	✓
Proactive Drift / SAR Risk	✗	✗	✓	✓
IMBL Current Guard	✗	✗	✗	✓
Safety-Constrained AI Decision	✗	✗	✗	✓
Multilingual Support	✓	✓	✓	✓

Mobile App UI/UX



LIVE PROTOTYPE
https://marvo-red.vercel.app

GITHUB REPOSITORY
https://github.com/akshatog/MARVO